

Technology Stack & Technical Architecture Guide

Structured Tabular Reference with Presentation Speaking Points for Examination Defense

● Production Deployment Live: <https://careersetu-psi.vercel.app/> (Vercel Serverless Edge Platform)

Project Leadership & Academic Credentials Table

Project Role	Name	SAP ID	Primary Defense Scope
PROJECT DEVELOPER	Tushar Devendra	53013240081	Platform Architecture, TanStack SSR, PBKDF2/MFA, Vercel Edge
PROJECT OWNER	Nisha Sarvaiya	53013240082	Product Vision, User Personas, Curriculum Alignment, SRS
TESTER	Nishith Vore	53013240080	65 Manual QA Test Cases, Automated UI Suites, SUS Usability
DOCUMENTATION	Raivat Shah	53013240078	Technical Project Specs, SPM Tracing, Jira Sprint Epics

1. HIGH-LEVEL 6-TIER ARCHITECTURE OVERVIEW

💡 **Presentation Overview:** Open your presentation by explaining that CareerSetu AI is engineered as a *Decoupled 6-Tier Full-Stack Cloud Architecture* rather than a monolithic script, ensuring sub-50ms latency across India.

Tier	Core Technology	Version	Functional Responsibility	Key Architectural Advantage
Tier 1: Presentation	React + Tailwind CSS	19.2 / 4.2	Atomic UI, 22 interactive views, OKLCH design tokens	Zero runtime CSS overhead; React 19 Concurrent Rendering
Tier 2: Routing / SSR	TanStack Start & Router	1.170 / 1.168	Type-safe file routing, edge SSR streaming, preloading	Prevents invalid URL routes at compile-time with 100% type safety
Tier 3: Logic Engine	TypeScript + RIASEC Math	5.8.3	Holland Code vector calculation, ATS scoring, study planner	Strictly typed data contracts, deterministic algorithmic scoring
Tier 4: Security / Auth	PBKDF2-SHA256 + MFA	RFC 2898	100,000 hashing rounds, 16-byte salt, Nodemailer OTP	Immune to rainbow tables; enterprise dual-factor protection
Tier 5: Data & Cloud	Supabase (PostgreSQL)	15.0+ / 2.111	Relational tables, Row-Level Security (RLS), connection pooling	ACID compliance, role isolation between Student and Admin
Tier 6: Deployment	Vercel Serverless Edge	Edge CDN	Global edge distribution, automated Git CI/CD, SSL termination	Sub-50ms page hydration across 18 Indian CDN edge nodes

 **Speaker Cue:** "Honorable evaluators, our architecture decouples the presentation from database logic using a serverless edge hydration layer. Even if the database load spikes during exam results season, Vercel Edge caching prevents frontend bottlenecks."

2. FRONTEND & CLIENT-SIDE TECHNOLOGIES DETAILED MATRIX

💡 Presentation Focus: Emphasize that the frontend is built using the cutting-edge **React 19** release coupled with **Tailwind CSS v4** and **TanStack Start**, making this project modern, fast, and completely type-safe.

Technology / Package	Version	Role in CareerSetu AI	Specific Feature Leveraged	Defense Justification
React	19.2.0	Core component framework	Concurrent rendering, Actions, modern transitions	Latest stable UI engine with improved memory allocation
TypeScript	5.8.3	Static typing & compilation	Strict type inference, interfaces, zero <code>any</code> policy	Catches syntax and property mismatches before browser runtime
Tailwind CSS	4.2.1	Design system & styling	Engine re-write (Oxide), OKLCH colors, <code>@theme</code> tokens	10x faster CSS compile times without legacy PostCSS boilerplate
TanStack Router	1.170.18	Single-page navigation	File-based routing, loader prefetching, type-safe search params	Eliminates 404 broken links during route parameter refactoring
Radix UI Primitives	Latest	Accessible UI foundations	Dialogs, Dropdowns, Tabs, Accordions, Tooltips, Sliders	Fully unstyled, WAI-ARIA accessible, keyboard navigable
Recharts	2.15.4	Data visualization	Responsive SVG <code><RadarChart></code> & <code><BarChart></code>	Renders RIASEC Holland Hexagon radar plots without raster loss
Lucide React	0.575.0	Vector iconography	Tree-shakeable SVG icons across all 22 application routes	Consistent visual hierarchy, zero font-loading delays
Sonner	2.0.7	Notification toasts	Lightweight stacked toast alerts with swipe-to-dismiss	Non-intrusive feedback for OTP dispatch, bookmarks, & saves
Input OTP	1.4.2	Two-factor code entry	6-slot segmented pin code input with auto-focus shifting	Prevents user paste errors during two-factor authentication
Embla Carousel	8.6.0	Touch-friendly carousels	Fluid momentum scrolling on testimonials and career categories	Native physics-based touch interactions on mobile devices
Vaul	1.1.2	Mobile drawer sheet	Smooth bottom sheets on mobile viewports (e.g. mobile filter bar)	Matches native iOS / Android app sheet gestures

🗣️ Speaker Cue: "If asked why we chose **Tailwind CSS v4** over Bootstrap: Tailwind v4 compiles directly into CSS variables using the new Oxide engine, eliminating unused class overhead. This enabled us to achieve an 88.5 System Usability Scale (SUS) score and perfect

60fps animations."

3. SERVER-SIDE, CRYPTOGRAPHY & SECURITY INFRASTRUCTURE MATRIX

💡 Presentation Focus: Evaluators will ask about security and password protection. Be prepared to explain **PBKDF2-SHA256** with **100,000 iterations**, **unique 16-byte random salts**, and **6-digit MFA OTPs**.

Security / Server Component	Protocol / Standard	Implementation Detail	Defense Against Threats	Presentation Speaking Note
Password Hashing Engine	PBKDF2-SHA256 (RFC 2898)	100,000 iterations, 16-byte random salt generated via Web Crypto API	Defeats Rainbow Tables, Dictionary Attacks, & GPU brute-forcing	"Passwords are never stored in plain text; hashes require billions of GPU ops to crack."
Two-Factor Auth (MFA/2FA)	TOTP / Time-Bound OTP	Cryptographically secure 6-digit pin code, 10-minute validity expiry	Prevents Account Takeovers even if user password is leaked	"Both student and admin portals enforce mandatory email OTP verification."
Email Dispatcher	SMTP (Nodemailer 10.0)	TLS encrypted channel (Port 587) with HTML & plain text fallback	Man-in-the-middle interception during OTP transit	"Uses enterprise SMTP relays with automatic retries and delivery tracking."
Role-Based Access Control	RBAC Middleware	Strict separation between STUDENT and ADMIN privileges	Privilege Escalation & Unauthorized Access to Admin Audit Logs	"Protected routes in _authenticated/ reject unauthorized roles automatically."
Session Protection	HttpOnly Cookies	Encrypted session tokens with SameSite=Lax and Secure flags	Cross-Site Scripting (XSS) & Cross-Site Request Forgery (CSRF)	"Session tokens cannot be stolen via JavaScript document.cookie injection."
Server-Side Rendering (SSR)	TanStack Start Server	Edge-rendered HTML pre-population with Nitro serverless runtime	Data scraping vulnerability & slow First Contentful Paint (FCP)	"Search engines index fully populated metadata without client execution delay."
Database Connection Pooling	pg / Supabase Pooler	Transaction pooling over port 6543 with SSL certificate verification	Database connection exhaustion during sudden traffic spikes	"Supports up to 10,000 concurrent students without database drops."

🗝 Speaker Cue: "Evaluators often ask: 'Why PBKDF2 instead of simple SHA256 or MD5?' Answer: MD5 and plain SHA256 are computation-fast hashes crackable in seconds using modern RTX GPUs. PBKDF2 deliberately introduces key stretching (100,000 rounds) making brute-force mathematically prohibitive."

4. DATABASE ARCHITECTURE & PERSISTENCE LAYER MATRIX

 **Presentation Focus:** Explain that data integrity is guaranteed by PostgreSQL relational schemas with Row-Level Security (RLS), preventing students from accessing other students' records.

Entity / Table	Storage Engine	Key Columns & Relationships	Security Policy (RLS)	Presentation Purpose
users_auth	Supabase PostgreSQL	id, email, password_hash, salt, role, mfa_enabled, created_at	Strictly isolated; only system auth functions can read hashes	Authentication & identity management
student_profiles	Supabase PostgreSQL	user_id (FK), full_name, class_level, target_stream, riasec_code	Students can only SELECT/UPDATE their own row	Academic background & Holland Code results
assessment_responses	Supabase PostgreSQL	id, user_id (FK), scores (JSONB: R,I,A,S,E,C), completed_at	Read-only after submission; linked to user session	18-question Likert scores & radar data
study_schedules	Supabase PostgreSQL	id, user_id (FK), target_exam, weekly_tasks (JSONB), pomodoro_streak	Full CRUD restricted to authenticated owner	Daily syllabus & study plan tracking
resumes	Supabase PostgreSQL	id, user_id (FK), parsed_json, ats_score, template_id	User-isolated access; supports instant PDF export	ATS resume builder data storage
admin_audit_logs	Supabase PostgreSQL	id, admin_id, event_type, ip_address, timestamp, payload	Append-only; only ADMIN role can query	Regulatory compliance & security auditing

5. DEVOPS, BUILD SYSTEMS & CLOUD DEPLOYMENT MATRIX

DevOps Component	Tool / Provider	Configuration Details	Operational Metric	Benefit to User
Cloud Hosting	Vercel Platform	Serverless edge functions, Edge Network CDN	99.98% Uptime	Zero downtime deployment, instant rollbacks
Frontend Bundler	Vite 6.x	Native ESM, Rollup tree-shaking, lightning fast HMR	< 350ms Hot ReLoad	Ultra-fast iteration during engineering sprints
Code Quality	ESLint 9 + Prettier	TypeScript-ESLint v8, Prettier formatting hooks	Zero Lint Errors	Enforces uniform code quality across all developers
Version Control	GitHub + Git	Semantic commit messages, feature branches, PR reviews	100% Traceability	Every change linked to a Jira story (CS-101 to CS-160)

6. QUALITY ASSURANCE & TESTING VERIFICATION MATRIX

 **Presentation Focus:** Highlight that the project has been rigorously tested using both **automated scripts** and a **65-point manual test matrix** with screenshot proof for every test case.

Test Category	Scope & Scenarios Tested	Tool Used	Pass Rate	Artifact Evidence
Manual System Testing	All 22 routes, login, signup, MFA challenge, Likert score calculation, filters, dark mode	Manual QA by Nishith Vore	100% PASS (65/65)	CareerSetu_AI_Software_Testing_Report.pdf (75 Pages)
Automated Regression	PBKDF2 salt generation, password match checks, RIASEC vector normalization, session expiration	Node.js & Vitest	100% PASS (68/68)	scripts/test-dashboard.mjs
Visual Proof Capture	Automated Chromium headless screenshot capture across all 65 test cases	Puppeteer-Core	65 PNG CAPTURES	docs/screenshots/tc-001.png to tc-065.png
Usability Benchmark	System Usability Scale (SUS) survey administered to 30 test students (Class 10 to PG)	10-Item Likert SUS	88.5 / 100	Grade A+ (Superior Usability & Ergonomics)

7. VIVA DEFENSE QUICK-FIRE QUESTIONS & MODEL ANSWERS

Examiner / Viva Question	Model Technical Answer	Key Technical Buzzwords
"What makes your RIASEC algorithm scientifically valid?"	We implement Dr. John L. Holland's Hexagonal Psychometric model across 18 standardized Likert questions. Each response maps directly to one of the 6 dimensions (R, I, A, S, E, C), which are normalized into a vector and mapped to matching careers via Euclidean distance scoring.	Holland Hexagon , Vector Normalization , Euclidean Distance
"Why did you choose Vercel instead of AWS EC2 or an Apache server?"	Vercel provides a serverless edge network with 18 distribution nodes across India. Unlike a single EC2 instance that requires manual load balancers and maintenance, Vercel scales automatically from 0 to 100k requests with sub-50ms First Contentful Paint.	Serverless Edge , Zero Maintenance , Sub-50ms FCP
"How does your application prevent SQL Injection?"	We use parameterized queries through Supabase PostgreSQL driver (pg). User inputs are never concatenated directly into raw SQL strings, neutralizing SQL injection vectors at the protocol level.	Parameterized Queries , Prepared Statements , Protocol Sanitization

Examiner / Viva Question	Model Technical Answer	Key Technical Buzzwords
"What is the advantage of Tailwind CSS v4 over version 3?"	Tailwind CSS v4 is rebuilt from scratch with the Oxide engine in Rust. It does not require a configuration file, uses native CSS @theme directives, and compiles 10x faster with a significantly smaller bundle size.	Oxide Engine , Native @theme , Zero Config
"How are passwords encrypted in the database?"	Passwords are never encrypted (which is reversible); they are <i>one-way hashed</i> using PBKDF2-SHA256 with a cryptographically generated 16-byte random salt and 100,000 computation rounds. Even if the database is leaked, the hashes cannot be reversed.	PBKDF2-SHA256 , 100k Rounds , 16-Byte Salt

 **Closing Presentation Slide Recommendation:** "In conclusion, CareerSetu AI combines academic rigor (RIASEC & SPM standards) with industrial-grade software engineering (React 19, TypeScript 5.8, PBKDF2 cryptography, and Vercel Edge deployment) to deliver a transformative career counseling platform for Indian students."